DEVOPS - 06-03-2026

Clone pannuna both layum same ip address and same name and same password tha varum so namma atha change pannanum

```
sudo hostnamectl set-hostname k8nworker1
```

```
sudo nano /etc/hosts
```

```
sudo nano /etc/netplan/01-netcfg.yaml
```

```
sudo netplan apply
```

```
sudo nano /etc/netplan/01-netcfg.yaml
```

```
network:
  version: 2
  renderer: networkd
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 172.20.10.5/24
      gateway4: 172.20.10.1
      nameservers:
        addresses: [8.8.8.8,8.8.4.4]
```

```
sudo netplan apply
```

```
sudo kubeadm init --pod-network-cidr=192.168.0.0/16
```

```
kubeadm version
```

```
lsmod | grep br_netfilter
```

```
echo br_netfilter | sudo tee /etc/modules-load.d/k8s.conf
```

```
cat /proc/sys/net/bridge/bridge-nf-call-iptables
```

```
echo "net.bridge.bridge-nf-call-iptables = 1" | sudo tee /etc/sysctl.d/k8s.conf
```

```
echo "net.ipv4.ip_forward = 1" | sudo tee -a /etc/sysctl.d/k8s.conf
```

```
sudo sysctl --system
```

```
sudo kubeadm reset -f
```

```
sudo kubeadm init --pod-network-cidr=192.168.0.0/16
```

```
Last login: Fri Mar  6 07:16:57 2026 from 10.119.248.109
lekha@Kube-master:~$ sudo kubeadm init --pod-network-cidr=192.168.0.0/16
[sudo] password for lekha:
W0306 08:46:10.008673    1178 version.go:104] could not fetch a Kubernetes v
ersion from the internet: unable to get URL "https://dl.k8s.io/release/stabl
e-1.txt": Get "https://dl.k8s.io/release/stable-1.txt": context deadline exc
eeded (Client.Timeout exceeded while awaiting headers)
W0306 08:46:10.009971    1178 version.go:105] falling back to the local clie
nt version: v1.29.15
```

```
Your Kubernetes control-plane has initialized successfully!

To start using your cluster, you need to run the following as a regular user
:

  mkdir -p $HOME/.kube
  sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
  sudo chown $(id -u):$(id -g) $HOME/.kube/config

Alternatively, if you are the root user, you can run:

  export KUBECONFIG=/etc/kubernetes/admin.conf

You should now deploy a pod network to the cluster.
Run "kubectl apply -f [podnetwork].yaml" with one of the options listed at:
  https://kubernetes.io/docs/concepts/cluster-administration/addons/

Then you can join any number of worker nodes by running the following on eac
h as root:

kubeadm join 10.119.248.17:6443 --token fha5y4.lpi5ka6ms00hbd26 \
        --discovery-token-ca-cert-hash sha256:394d543aceefd30b0d77026e08c94c
e53ed629bd8a0a50fee4b9ded34cddf22b
lekha@Kube-master:~$
```

mkdir -p $HOME/.kube

sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config

sudo chown $(id -u):$(id -g) $HOME/.kube/config

kubectl get nodes

```
lekha@Kube-master:~$ mkdir -p $HOME/.kube
lekha@Kube-master:~$ sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/confi
g
lekha@Kube-master:~$ sudo chown $(id -u):$(id -g) $HOME/.kube/config
lekha@Kube-master:~$ sudo -i
```

```
lekha@Kube-master:~$ sudo swapoff -a
lekha@Kube-master:~$ sudo -i
root@Kube-master:~# kubectl get nodes
NAME          STATUS     ROLES           AGE     VERSION
kube-master   NotReady   control-plane   8m42s   v1.29.15
root@Kube-master:~#
```

```
lekha@Kube-master:~$ sudo nano /etc/fstab
lekha@Kube-master:~$ sudo nano /etc/fstab
lekha@Kube-master:~$ systemctl status containerd
● containerd.service - containerd container runtime
     Loaded: loaded (/usr/lib/systemd/system/containerd.service; enabled; p>
     Active: active (running) since Fri 2026-03-06 08:42:11 UTC; 30min ago
       Docs: https://containerd.io
   Main PID: 654 (containerd)
      Tasks: 74
     Memory: 924.1M (peak: 937.1M)
        CPU: 3min 22.720s
     CGroup: /system.slice/containerd.service
             ├─ 654 /usr/bin/containerd
             ├─6101 /usr/bin/containerd-shim-runc-v2 -namespace k8s.io -id >
             ├─6203 /usr/bin/containerd-shim-runc-v2 -namespace k8s.io -id >
             ├─6303 /usr/bin/containerd-shim-runc-v2 -namespace k8s.io -id >
             ├─6578 /usr/bin/containerd-shim-runc-v2 -namespace k8s.io -id >
             └─6683 /usr/bin/containerd-shim-runc-v2 -namespace k8s.io -id >

Warning: some journal files were not opened due to insufficient permissions.
lines 1-17/17 (END)
```

```
lekha@Kube-master:~$ systemctl status kubelet
● kubelet.service - kubelet: The Kubernetes Node Agent
     Loaded: loaded (/usr/lib/systemd/system/kubelet.service; enabled; pres>
    Drop-In: /usr/lib/systemd/system/kubelet.service.d
             └─10-kubeadm.conf
     Active: active (running) since Fri 2026-03-06 09:00:26 UTC; 13min ago
       Docs: https://kubernetes.io/docs/
   Main PID: 2447 (kubelet)
      Tasks: 12 (limit: 2265)
     Memory: 33.4M (peak: 35.5M)
        CPU: 47.648s
     CGroup: /system.slice/kubelet.service
             └─2447 /usr/bin/kubelet --bootstrap-kubeconfig=/etc/kubernetes>
lines 1-12/12 (END)
```

Need to solve the below error

```
lekha@Kube-master:~$ sudo crictl ps
WARN[0000] runtime connect using default endpoints: [unix:///var/run/dockers
him.sock unix:///run/containerd/containerd.sock unix:///run/crio/crio.sock u
nix:///var/run/cri-dockerd.sock]. As the default settings are now deprecated
, you should set the endpoint instead.
ERRO[0000] validate service connection: validate CRI v1 runtime API for endp
oint "unix:///var/run/dockershim.sock": rpc error: code = Unavailable desc =
 connection error: desc = "transport: Error while dialing: dial unix /var/ru
n/dockershim.sock: connect: no such file or directory"
WARN[0000] image connect using default endpoints: [unix:///var/run/dockershi
m.sock unix:///run/containerd/containerd.sock unix:///run/crio/crio.sock uni
x:///var/run/cri-dockerd.sock]. As the default settings are now deprecated,
you should set the endpoint instead.
ERRO[0000] validate service connection: validate CRI v1 image API for endpoi
nt "unix:///var/run/dockershim.sock": rpc error: code = Unavailable desc = c
onnection error: desc = "transport: Error while dialing: dial unix /var/run/
dockershim.sock: connect: no such file or directory"
CONTAINER           IMAGE               CREATED             STATE
    NAME                      ATTEMPT             POD ID              POD
f89df91987da1       b0cdcf76ac8e9       3 minutes ago       Running
    kube-controller-manager   5                   5019577482ee8       kube-c
ontroller-manager-kube-master
lekha@Kube-master:~$
```

Overall commands seen today:

1. sudo hostnamectl set-hostname k8nworker1
2. sudo nano /etc/hosts
3. sudo nano /etc/netplan/01-netcfg.yaml
4. sudo netplan apply
5. sudo kubeadm init --pod-network-cidr=192.168.0.0/16
6. kubeadm version
7. lsmod | grep br_netfilter
8. sudo modprobe br_netfilter
9. echo br_netfilter | sudo tee /etc/modules-load.d/k8s.conf
10. cat /proc/sys/net/bridge/bridge-nf-call-iptables
11. echo "net.bridge.bridge-nf-call-iptables = 1" | sudo tee /etc/sysctl.d/k8s.conf
12. echo "net.ipv4.ip_forward = 1" | sudo tee -a /etc/sysctl.d/k8s.conf
13. sudo sysctl --system
14. sudo kubeadm reset -f
15. sudo kubeadm init --pod-network-cidr=192.168.0.0/16
16. mkdir -p $HOME/.kube
17. sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
18. sudo chown $(id -u):$(id -g) $HOME/.kube/config
19. kubectl get nodes
20. sudo nano /etc/fstab
21. systemctl status containerd
22. systemctl status kubelet
23. sudo crictl ps